

AAKARSH ANAND

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Google Scholar · GitHub · LinkedIn

Machine learning researcher working on foundation models and statistical genetics for health data

EDUCATION

Ph.D., Computer Science, UCLA · Major field: AI · Advisor: Sriram Sankararaman 2024 – 2029 (expected)
M.S., Computer Science, UCLA (earned during the Ph.D.) 2026
B.S., Computer Science, UCLA 2020 – 2024

RESEARCH EXPERIENCE

Graduate Student Researcher, Sankararaman Lab, UCLA (undergraduate researcher 2022–24) Jan 2022 – present

Foundation models for wearable sensors · with Yuzhe Yang's lab

- Co-first author of **Inertia-1** (NeurIPS 2026). Pretrained motion foundation models (5M–100M params, 6 architectures) on 18.2M hours of accelerometry from 115K+ people and compared data, model, and training choices on 15 datasets.
- Designed and ran the sensor-fusion study, which found that multi-sensor models consistently beat single-sensor ones.
- Co-first author of the follow-up journal paper (in prep.) on diagnosis, prognosis, and disease progression in five cohorts, including the genetics of the learned embeddings and early work pairing them with LLMs.

Deep learning for genetics

- Developing a training loss that makes a network's learned trait heritable and genetically correlated with disease, so a GWAS on it can find loci the original measurement misses. It works with any architecture.

Statistical genetics at biobank scale

- Leading a genome-wide test for gene–gene interactions in rare disease, supported by an Anthropic grant.
- Leading a single-cell model of how genetic effects on gene expression shift across cell types and states.
- Co-leading an atlas of genetic correlation across 3,570 UK Biobank trait pairs and 53 tissues.
- Ran simulations (calibration, power, runtime) and real-data analyses for the papers below.

PUBLICATIONS

* equal contribution

- Inertia-1: An Open Exploration of Wearable Motion Foundation Models
Z. Xu*, **A. Anand***, S. Jiang*, C. Zhuang, Z. Shuai, S. Sankararaman, Y. Yang · **NeurIPS 2026** · code
- A biobank-scale method for learning modulators of gene-environment interaction underlying human complex traits from multiple environmental exposures
Z. Liu, A. Ramteke, **A. Anand**, A. Gorla, M. Jeong, S. Sankararaman · **Genome Research 2026** · selected from RECOMB 2026
- A biobank-scale test of marginal epistasis reveals genome-wide signals of polygenic interaction effects
B. Fu, A. Pazokitoroudi, Z. Shi, A. Kar, A. Xue, **A. Anand**, P. Anand, et al. · **Nature Genetics 2025**
- A scalable adaptive quadratic kernel method for interpretable epistasis analysis in complex traits
B. Fu*, P. Anand*, **A. Anand***, J. Mefford, S. Sankararaman · **Genome Research 2024** · selected from RECOMB 2024
- Disentangling phenotype scale, amplification, and effect heterogeneity underlying polygenic GxE architecture of complex traits
Z. Liu, **A. Anand**, X. Qie, M. Costantino, M. Jeong, A. Gorla, N. Zaitlen, A. Dahl, S. Sankararaman · submitted to **Nature Genetics**
- Comprehensive gene heritability estimation reveals the genetic architecture of rare coding variants underlying complex traits
Z. Liu, B. Fu, M. Jeong, P. Anand, **A. Anand**, et al. · bioRxiv 2025 · in revision at **Nature Genetics**

AWARDS AND FUNDING

Anthropic AI for Science Grant, Rare Disease Research 2026
NIH T32 Predoctoral Fellowship, UCLA Genomic Analysis Training Program 2026 – 2027
Warren Alpert Computational Biology and AI Network Fellowship, UCLA 2024

TEACHING

Teaching Assistant, UCLA 2025 – 2026

- AI for Computational Genomics (Spring 2025, ~120 students). Co-wrote three new assignments for its AI redesign.
- Machine Learning Applications in Genetics (Spring 2026) and Machine Learning Applications in Biomedicine (Winter 2025).

EARLIER EXPERIENCE

Software Engineering Intern, LabCorp · Snakemake pipelines for high-volume sequencing data Jun – Sep 2023
Research Assistant, Boutros Lab, UCLA · cancer genomics pipelines, co-author of Metapipeline-DNA 2021 – 2022

SKILLS

Machine learning: PyTorch, scikit-learn, Transformers, self-supervised pretraining, time-series and wearable sensor data
Statistics and genetics: variance-component models, heritability and genetic correlation, GWAS, LDSC, single-cell genomics
Engineering: Python, NumPy/pandas, Weights & Biases, SLURM, Claude Code · **Data:** UK Biobank, NHANES, OneK1K, GTEx